

The Sugar Vault

An Arduino controlled candy & chocolate dispensing machine

PARTS+BOM

Start Date: 5/18/2026

End Date: 6/28/2026

Software Used: Google Docs, SOLIDWORKS, draw.io, Arduino IDE, GitHub, Bambu Studio

Project GitHub: <https://github.com/cchapman9x/the-sugar-vault>

Project Creator:

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Parts Required		
Part	Qty	Purpose
Arduino Nano	x1	Main microcontroller used to control the dispenser
A4988 Stepper Motor Driver	x1	Controls the NEMA 17 stepper motor
NEMA 17 Stepper Motor	x1	Rotates the dispensing wheel
12V 2A Power Supply	x1	Main external power source for the system
DC-DC Buck Converter	x1	Steps the 12V supply down to a safe voltage for the Arduino
Momentary Push Button	x1	User input used to start the dispensing cycle
Green LED	x1	Visual indicator that turns on while dispensing
2.2kΩ Resistor	x1	Limits current for the LED
220μF 25V Electrolytic Capacitor	x1	Helps smooth power near the stepper motor driver
Red & White Filament	As Needed	Used to create the 3D printed parts
12" Clear Acrylic Tube Ø100mm / 4"	x1	Used to store the candy and acts as the hopper of the system

Jumper Wires / Hookup Wires	As Needed	Used to connect the circuit components
Screw Terminals / Connectors	As Needed	Used for power and wiring connections
Breadboard-Style Perfboard / Prototyping Board	x2	Used to mount and organize the circuit components in a more permanent layout than a standard breadboard
M3x8mm Threaded Heat-Set Inserts	x16	Used for assembling the machine together
M3x10mm Screws	x20	Used for assembling the machine and motor

Tools Required	
Tool	Purpose
3D Printer	Printed the mechanical enclosure and dispensing parts
Wire Strippers	Removed insulation from wires
Wire Cutters	Cut wires to the correct length
Pliers	Used to remove support material and grip small parts/wires during assembly
Small Screwdriver Set	Tightened terminals, screws, and electronics connections
Soldering Iron	Used for permanent or semi-permanent electrical connections
Solder	Used with the soldering iron to connect wires/components
Multimeter	Checked voltage, continuity, and wiring connections
Heat Shrink	Used to insulate and protect soldered wire connections
Calipers	Used for part measuring